



King County Housing EDA — Client Presentation

Client & goal

- **Client:** *Jennifer Montgomery* (home buyer)
- **Goal:** Identify where “luxury” value shows up in King County using grade, condition, geographical data, price and size
- **Dataset:** King County house sales (features + sale price)

Project overview

- **What we did:** Exploratory Data Analysis (EDA) on house attributes vs. price
- **Main question:** Which combinations of features reliably predict the highest prices?
- **How to read this deck:** Each hypothesis shows (1) the idea, (2) the plot(s), (3) what it means for Jennifer

Dataset at a glance

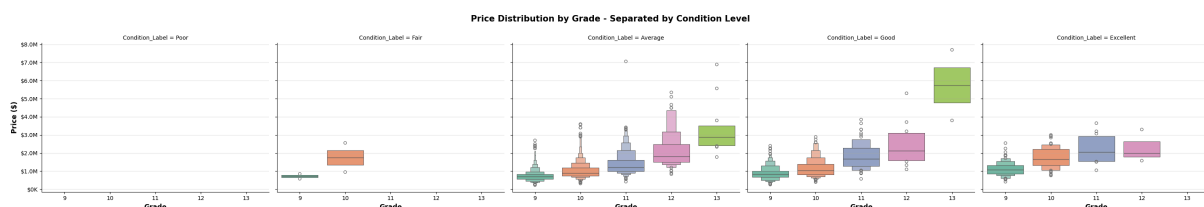
- **Target variable:** 🏠 Price, 📏 Grades, 💧 Waterfront, 🏡 Condition
- **Key features used in hypotheses:**
 - Grade (overall construction and design quality)
 - Condition (overall state of the property)
 - Geographical location 📍
 - Living area (sqft living)
 - Price (💰)
- *Other datasets which are considered but not necessarily used in understanding the data (Riverfront, Year renovated, Year built).*

Hypothesis 1 — *Grade & Condition Premium*

- **Claim:** *High grade (10–13) + good condition (4–5) produces the highest prices*
Broader market availability
- **Also true:** These homes are available across more of the county, not only in one area

EDA for hypothesis 1:

- Conditions by Grade × Price



Hypothesis 1 — *Findings*

- **Result:**

- We found that better houses are in condition range Good - Excellent and grade higher than 10 are premium.
- We can target these areas as per the Client's requirements.

• **Client takeaway:** Jennifer can search broadly, but should prioritize **grade 10+** and **condition 4-5**

• **Recommendation:** Use grade and condition as "must-have" filters before location fine-tuning

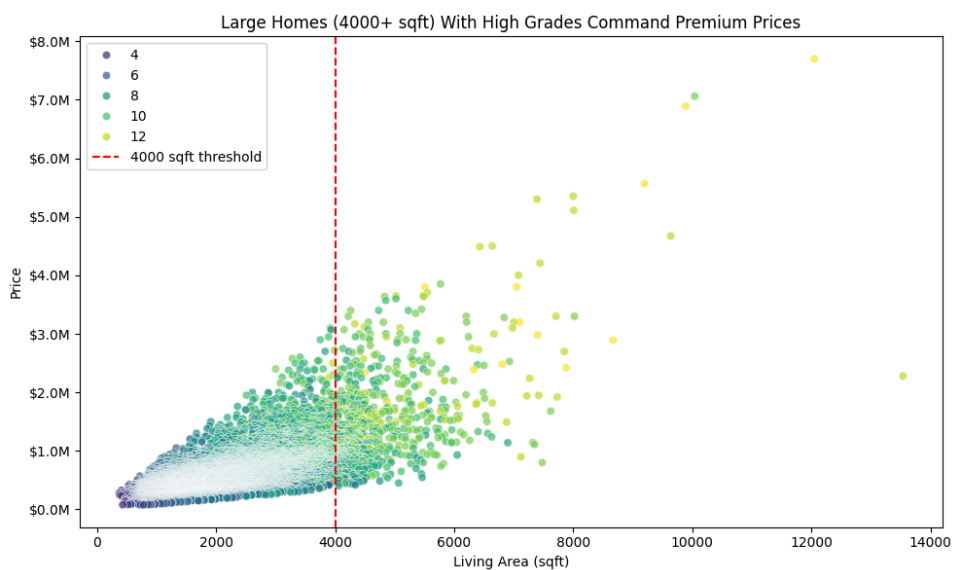
Hypothesis 2 — Size + Grade + Locality= Premium

• **Claim:** Large homes (4000+ sqft) with high grade command **premium prices**

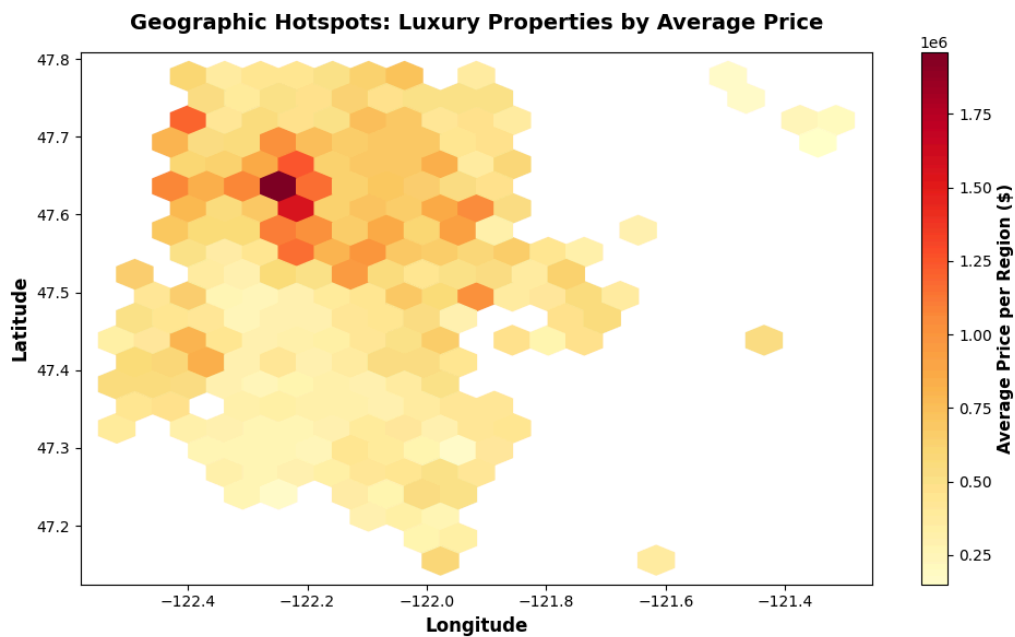
• **Why it matters:** Spacious + high quality aligns with "luxury appeal" and resale demand

EDA for hypothesis 2:

- Price vs. sqft living (scattered with grade)
- Price distribution for 4000+ sqft vs. below



- Price vs Geographical data (Premium locations)



Hypothesis 2 — Findings

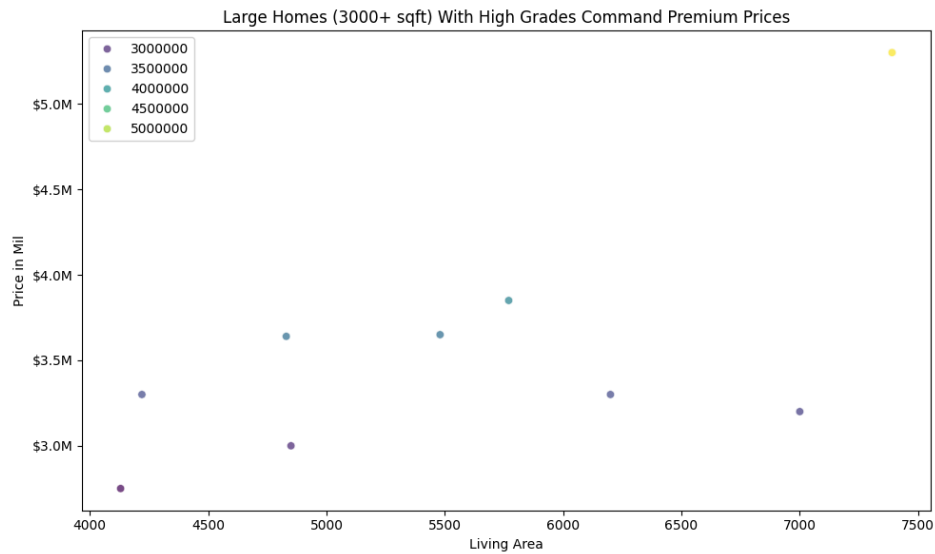
- **Result:**
 - Higher grade observed above 4000 sqft living space with higher price range.
 - Good locality found between Lat > 47.5 and Long < -122.0
- **Client takeaway:** For standout properties, size helps, but size alone is not enough without high grade and good location
- **Recommendation:** Shortlist homes that meet **4000+ sqft** and **grade 10+** first, then assess location and features

Hypothesis 3 — “Luxury Package” (Multi-factor)

- **Definition:**
 - 🏡 Grade 10+
 - 🏠 Condition 4+
 - 📏 Size 4000+ sqft
 - 📍 Latitude > 47.5 & Longitude < -122.0
 - 💧 Waterfront = Yes
- **Claim:** This combination represents the **true luxury segment** and the hottest resale items

EDA for hypothesis 3:

- Distribution of prices for Luxury Package
- Count of homes meeting the package criteria



Hypothesis 3 — Findings

- **Luxury segment summary:**
 - 💰 **Average price: \$3.55M**
 - **Typical range (IQR): \$2.75M–\$5.3M**
 - **Availability:** 9 homes
- **Client takeaway:** If Jennifer wants “true luxury,” this package is the clearest filter and supports confident resale expectations

List	house_id	Price 💰	yr_built	grade	condition	sqft_living	yr_renov
0	7558700030	\$5.3M	1991	12	4	7390.000	NA
1	8106100105	\$3.85M	1980	11	4	5770.000	NA
2	4217402115	\$3.65M	1936	11	5	5480.000	NA
3	2425049063	\$3.64M	1990	11	4	4830.000	NA
4	3625059043	\$3.3M	1989	11	4	6200.000	NA
5	3625059152	\$3.3M	1958	11	4	4220.000	1987
6	624069108	\$3.2M	1991	12	4	7000.000	NA
7	3761100045	\$3.0M	1989	10	5	4850.000	NA
8	624069035	\$2.75M	1993	10	4	4130.000	NA

List of houses as per years renovated falling in our selected region comprising the conditions

List	house_id	Price 💰	yr_built	grade	condition	sqft_living	yr_renov
0	1924059029	\$4.67M	1983	12	3	9640.000	2009
1	4114601570	\$3.6M	1941	10	3	5020.000	2002
2	9808700762	\$7.0M	1940	11	3	10040.000	2001
3	9201300050	\$1.85M	1963	9	3	2800.000	2001
4	7936500172	\$1.18M	1937	8	3	1970.000	1998
5	7352200100	\$1.36M	1931	8	3	2620.000	1991
6	623059016	\$1.1M	1956	8	3	3190.000	1991

Practical buying guidance (how to use this)

- **Step 1:** Decide if “Luxury Package” is a hard requirement or a nice-to-have
 - **Step 2:** If budget is tight, relax one constraint (usually size first) and keep **grade** as the anchor
 - **Step 3:** Verify with comps and neighbourhood context for finalist homes
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